

# BENJAMIN HAN

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## PROFESSIONAL SUMMARY

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I am a biomedical engineer specializing in AI-powered systems that integrate multimodal healthcare data across literature, structured databases, and medical imaging. My work involves developing and evaluating scalable machine learning and LLM-based algorithms to transform complex medical data into actionable and accessible information for clinicians.

## EDUCATION

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**University of Washington** | Seattle, WA  
**Master of Applied Bioengineering** | GPA: 3.94  
Graduation Date – June 2026

**University of Washington** | Seattle, WA | Annual Dean's List  
**Bachelor of Science – Bioengineering: Data Science** | GPA: 3.47  
**Minor: Applied Mathematics**

## SKILLS AND QUALIFICATIONS

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**Programming Languages:** Python, MATLAB, R, SQL, Java, Dart, Visual Basic for Applications (VBA), HTML, CSS  
**AI/ML:** PyTorch, TensorFlow, scikit-learn, RAG pipelines, synthetic query generation, prompt engineering, LLM evaluation  
**Engineering and Design Software:** Fusion 360, AutoCAD, LabVIEW, COMSOL Multiphysics, Onshape, SolidWorks

## TECHNICAL PROJECTS

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### **WiseFind: Medical AI Platform for Neurotrauma Assessment**

*September 2025 – Present*

University of Washington | Seattle, WA

- Designing and validating a Python data processing pipeline for a Retrieval-Augmented Generation system for handling neurotrauma queries, securing \$55,000 in funding from ITHS and the UW Neurosurgery Department
- Developed a synthetic query generation system that expanded LLM evaluation coverage from 20 manually written prompts to 400+ clinician-style queries, simulating realistic usage conditions
- Ensured project continuity and onboarding efficiency for new project members by maintaining detailed documentation and standard operating procedures (SOPs) for ongoing AI platform development

### **WiseSpine: Image-Focused AI Agent for Spine MR/CT Analysis**

*November 2025 – Present*

University of Washington | Seattle, WA

- Developed a Python-based spine reconstruction pipeline for automated Cobb angle measurement using 481 spine X-ray images and three deep learning segmentation models
- Built a segmentation benchmarking system for medical imaging using spatial analysis and quantitative metrics
- Accelerated scalable AI infrastructure design by collaborating with AWS engineers on pipeline development

### **CareChorus: Patient-Centered Care Navigation App**

*March 2025 – Present*

University of Washington | Seattle, WA

- Validated an AI-driven mobile application designed for patient treatment navigation across healthcare systems
- Designed an evaluation pipeline in Dart to benchmark classification and data extraction of medical documents
- Supported product development by collaborating with stakeholders on feature design and implementation

## WORK EXPERIENCE

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### **Quality Assurance Intern**

*September 2024 – Present*

Bio-Therapeutic | Seattle, WA

- Increased operational visibility across 1000+ purchasing orders and supply records by developing three interactive Tableau dashboards for inventory and procurement tracking
- Reduced manual data entry effort and improved workflow by automating customer service logging workbooks
- Supported quality management reporting by collecting and analyzing statistical client data for internal review
- Conducted surveillance audits to ensure compliance with the Bio-Therapeutic Quality Management System

### **Graduate/Undergraduate Lead Teaching Assistant**

*January 2024 – Present*

UW Bioengineering Department | Seattle, WA

- Strengthened student understanding of MATLAB programming and bioengineering design concepts by working with instructors to develop course materials, revise online modules, and lead undergraduate lectures
- Supported academic success through holding 150+ office hours and mentoring eight student engineering teams

### **SummerWorks Engineering Intern**

*June 2023 – August 2023*

City of Portland Bureau of Environmental Services | Portland, OR

- Improved accessibility of engineering records by designing a VBA-based tool managing 500+ piping data entries
- Reduced manual engineering calculations by implementing Excel-based automation for pump curve analysis
- Streamlined documentation workflows by annotating pump station schematics for infrastructure records